

Augmented Reality

What is Augmented Reality? Augmented reality (AR) is an enhanced and interactive version of the real-world environment achieved through digital visuals, sounds, and other sensory effects using holographic technology. AR encompasses three key features: a combination of digital and physical worlds, real-time interactions, and precise 3D rendering of virtual and real elements.

AR provides a better way to design, format, and deliver usable instructions by overlaying digital content onto real-world work environments. When a company understands what AR is and how to leverage it effectively, everyone can work remotely and collaborate efficiently.

Augmented Reality vs. Virtual Reality vs. Mixed Reality vs. Extended Reality

With the recent growth of virtual reality types, keeping up can be challenging due to their subtle differences. Digital reality types include:

Augmented Reality (AR)—designed to overlay digital elements onto real-world scenes with limited interaction.

Virtual Reality (VR)—immersive experiences that help isolate users from the real world, typically through a headset and headphones designed for such activities. Mixed Reality (MR)—combining augmented reality (AR) and VR elements so that digital objects can interact with the real world—means that businesses can design objects that are embedded in a real-world environment.

Extended Reality (XR)—covers all types of technologies that enhance our senses, including the three types mentioned above.

Types of Augmented Reality

When deciding which type of AR technology you need for your business, you'll first need to decide which type of AR you want to use. There are two main types of AR: tag-based and tagless. Choosing one of these AR types will determine how you will be able to display your images and information.

Tag-based AR is created using image recognition to identify objects that are already programmed into an AR device or application. When these objects are placed in the view as reference points, they help the AR device determine the camera's position and orientation. This is generally achieved by switching the camera to grayscale and detecting a tag to compare it with all the other tags in its database. Once your device finds a match, it uses that data to determine the mathematical shape and place the AR image in the correct location.

Tagless augmented reality is the most complex because there's no point of focus for your device. Therefore, your device must recognize objects as they appear in the view. Using a recognition algorithm, the device will search for similar colors, patterns, and features to identify the object. Then, using time information, an accelerometer, GPS, and a compass, it will orient itself and use the camera to overlay an image of whatever you want in your real-world surroundings.

How Augmented Reality Works

Augmented reality creates an immersive experience for all its users. While the most common forms of augmented reality are through glasses or a camera lens, interest in AR is growing, and companies are offering more types of lenses and devices through the market. There are five key components of augmented reality:

Artificial Intelligence (AI). Most AR solutions require AI to function, allowing users to complete actions using voice prompts. AI can also help process information for the AR application.

Augmented Reality Software. These are the tools and applications used to access augmented reality. Some companies can create their own proprietary AR software.

Processing. You'll need processing power for AR technology to work, generally by leveraging your device's internal operating system.

Lenses. You'll need an image processing system or lens to display content or images. The better the screen quality, the more realistic your image will appear.

Sensors. Augmented reality systems need to gather data about their environment to align the real and digital worlds. When a camera captures information, it sends it through software for processing.

Integrating AR into Employee Training and Education

In the workplace, adding augmented reality to your processes and procedures can enhance the learning and understanding benefits for your employees. AR training is an educational experience delivered through software on AR devices to help employees acquire important professional skills. This type of training experience can be run anytime, anywhere with the appropriate software.

Augmented reality can also help guide and support employees regardless of their location, leading to better collaboration and safer working conditions in your field. By enhancing traditional learning methods, this approach can provide more information for a deeper understanding. Some ways your team could use augmented reality include:

Performance support

Learning and training modules

New employee onboarding

On-demand training opportunities

Customer service and experience